11.3 A GAME THEORY APPROACH TO H_∞ FILTERING

The H_∞ solution that we present in this section was originally developed by Ravi Banavar [Ban92] and is further discussed in [She95, She97]. Suppose we have the standard linear discrete-time system

$$\begin{aligned} x_{k+1} &= F_k x_k + w_k \\ y_k &= H_k x_k + v_k \end{aligned} \tag{11.34}$$

where w_k and v_k are noise terms. These noise terms may be random with possibly unknown statistics, or they may be deterministic. They may have a nonzero mean. Our goal is to estimate a linear combination of the state. That is, we want to estimate z_k, which is given by

$$z_k = L_k x_k \tag{11.35}$$

where L_k is a user-defined matrix (assumed to be full rank). If we want to directly estimate x_k (as in the Kalman filter) then we set $L_k = I$. But in general we may only be interested in certain linear combinations of the state. Our estimate of z_k is denoted $\hat{z}_k$, and our estimate of the state at time 0 is denoted $\hat{x}_0$. We want to estimate z_k based on measurements up to and including time $(N-1)$. In the game theory approach to H_∞ filtering we define the following cost function:

$$J_1 = \frac{\sum_{k=0}^{N-1} ||z_k - \hat{z}_k||_{S_k}^2}{||x_0 - \hat{x}_0||_{P_0^{-1}}^2 + \sum_{k=0}^{N-1}\left(||w_k||_{Q_k^{-1}}^2 + ||v_k||_{R_k^{-1}}^2\right)} \tag{11.36}$$

Our goal as engineers is to find an estimate $\hat{z}_k$ that minimizes J_1. Nature's goal as our adversary is to find disturbances w_k and v_k, and the initial state x_0, to maximize J_1. Nature's ultimate goal is to maximize the estimation error $(z_k - \hat{z}_k)$. The way that nature maximizes $(z_k - \hat{z}_k)$ is by a clever choice of w_k, v_k, and x_0. Nature could maximize $(z_k - \hat{z}_k)$ by simply using infinite magnitudes for w_k, v_k, and x_0, but this would not make the game fair. That is why we define J_1 with $(x_0 - \hat{x}_0)$, w_k, and v_k in the denominator. If nature uses large magnitudes for w_k, v_k, and x_0 then $(z_k - \hat{z}_k)$ will be large, but J_1 may not be large because of the denominator. The form of J_1 prevents nature from using brute force to maximize $(z_k - \hat{z}_k)$. Instead, nature must try to be clever in its choice of w_k, v_k, and x_0 as it tries to maximize $(z_k - \hat{z}_k)$. Likewise, we as engineers must be clever in finding an estimation strategy to minimize $(z_k - \hat{z}_k)$.

This discussion highlights a fundamental difference in the philosophy of the Kalman filter and the H_∞ filter. In Kalman filtering, nature is assumed to be indifferent. The pdf of the noise is given. We (as filter designers) know the pdf of the noise and can use that knowledge to obtain a statistically optimal state estimate. But nature cannot change the pdf to degrade our state estimate. In H_∞ filtering, nature is assumed to be perverse and actively seeks to degrade our state estimate as much as possible. Intuition and experience seem to indicate that neither of these extreme viewpoints of nature is entirely correct, but reality probably lies somewhere in the middle.[1]

[1]Nevertheless, it is advisable to remember the *principle of perversity of inanimate objects* [Bar01, p. 96] – for instance, when dropping a piece of buttered toast on the floor, the probability is significantly more than 50% that the toast will land buttered-side down.

P_0, Q_k, R_k, and S_k in Equation (11.36) are symmetric positive definite matrices chosen by the engineer based on the specific problem. For example, if the user is particularly interested in obtaining an accurate estimate of the third element of z_k, then $S_k(3,3)$ should be chosen to be large relative to the other elements of S_k. If the user knows *a priori* that the second element of the w_k disturbance is small, then $Q_k(2,2)$ should be chosen to be small relative to the other elements of Q_k. In this way, we see that P_0, Q_k, and R_k are analogous to those same quantities in the Kalman filter, if those quantities are known. That is, suppose that we know that the initial estimation error, the process noise, and the measurement noise are zero-mean. Further suppose that we know their covariances. Then we should use those quantities for P_0, Q_k, and R_k in the H$_\infty$ estimation problem. In the Kalman filter, there is no analogy to the S_k matrix given in Equation (11.36). The Kalman filter minimizes the S_k-weighted sum of estimation-error variances for all positive definite S_k matrices (see Section 5.2). But in the H$_\infty$ filter, we will see that the choice of S_k affects the filter gain.

The direct minimization of J_1 is not tractable, so instead we choose a performance bound and seek an estimation strategy that satisfies the threshold. That is, we will try to find an estimate $\hat{z}_k$ that results in

$$J_1 < \frac{1}{\theta} \tag{11.37}$$

where θ is our user-specified performance bound. Rearranging this equation results in

$$\begin{aligned} J &= \frac{-1}{\theta}||x_0 - \hat{x}_0||^2_{P_0^{-1}} + \sum_{k=0}^{N-1}\left[||z_k - \hat{z}_k||^2_{S_k} - \frac{1}{\theta}\left(||w_k||^2_{Q_k^{-1}} + ||v_k||^2_{R_k^{-1}}\right)\right] \\ &< 1 \end{aligned} \tag{11.38}$$

where J is defined by the above equation. The minimax problem becomes

$$J^* = \min_{\hat{z}_k} \max_{w_k, v_k, x_0} J \tag{11.39}$$

Since $z_k = L_k x_k$, we naturally choose $\hat{z}_k = L_k \hat{x}_k$ and try to find the $\hat{x}_k$ that minimizes J. This gives us the problem

$$J^* = \min_{\hat{x}_k} \max_{w_k, v_k, x_0} J \tag{11.40}$$

Nature is choosing x_0, w_k, and v_k to maximize J. But x_0, w_k, and v_k completely determine y_k, so we can replace the v_k in the minimax problem with y_k. We therefore have

$$J^* = \min_{\hat{x}_k} \max_{w_k, y_k, x_0} J \tag{11.41}$$

Since $y_k = H_k x_k + v_k$, we see that $v_k = y_k - H_k x_k$ and

$$||v_k||^2_{R_k^{-1}} = ||y_k - H_k x_k||^2_{R_k^{-1}} \tag{11.42}$$

Since $z_k = L_k x_k$ and $\hat{z}_k = L_k \hat{x}_k$, we see that

$$\begin{aligned} ||z_k - \hat{z}_k||^2_{S_k} &= (z_k - \hat{z}_k)^T S_k (z_k - \hat{z}_k) \\ &= (x_k - \hat{x}_k)^T L_k^T S_k L_k (x_k - \hat{x}_k) \\ &= ||x_k - \hat{x}_k||^2_{\bar{S}_k} \end{aligned} \tag{11.43}$$

where $\bar{S}_k$ is defined as

$$\bar{S}_k = L_k^T S_k L_k \tag{11.44}$$

We substitute these results in Equation (11.38) to obtain

$$\begin{aligned} J &= \frac{-1}{\theta}||x_0 - \hat{x}_0||^2_{P_0^{-1}} + \sum_{k=0}^{N-1} \left[||x_k - \hat{x}_k||^2_{\bar{S}_k} - \frac{1}{\theta}\left(||w_k||^2_{Q_k^{-1}} + ||y_k - H_k x_k||^2_{R_k^{-1}} \right) \right] \\ &= \psi(x_0) + \sum_{k=0}^{N-1} \mathcal{L}_k \end{aligned} \tag{11.45}$$

where $\psi(x_0)$ and $\mathcal{L}_k$ are defined by the above equation. To solve the minimax problem, we will first find a stationary point of J with respect to x_0 and w_k, and then we will find a stationary point of J with respect to $\hat{x}_k$ and y_k.

11.3.1 Stationarity with respect to x_0 and w_k

The problem in this section is to maximize $J = \psi(x_0) + \sum_{k=0}^{N-1} \mathcal{L}_k$ (subject to the constraint $x_{k+1} = F_k x_k + w_k$) with respect to x_0 and w_k. This is the dynamic constrained optimization problem that we solved in Section 11.2.3. The Hamiltonian for this problem is defined as

$$\mathcal{H}_k = \mathcal{L}_k + \frac{2\lambda_{k+1}^T}{\theta}(F_k x_k + w_k) \tag{11.46}$$

where $2\lambda_{k+1}/\theta$ is the time-varying Lagrange multiplier that must be computed $(k = 0, \cdots, N-1)$. Note that we have defined the Lagrange multiplier as $2\lambda_{k+1}/\theta$ instead of λ_{k+1}. This does not change the solution to the problem, it simply scales the Lagrange multiplier (in hindsight) by a constant to make the ensuing math more straightforward. From Equation (11.33) we know that the constrained stationary point of J (with respect to x_0 and w_k) is solved by the following four equations:

$$\begin{aligned} \frac{2\lambda_0^T}{\theta} + \frac{\partial \psi_0}{\partial x_0} &= 0 \\ \frac{2\lambda_N^T}{\theta} &= 0 \\ \frac{\partial \mathcal{H}_k}{\partial w_k} &= 0 \\ \frac{2\lambda_k^T}{\theta} &= \frac{\partial \mathcal{H}_k}{\partial x_k} \end{aligned} \tag{11.47}$$

From the first expression in the above equation we obtain

$$\begin{aligned} \frac{2\lambda_0}{\theta} - \frac{2}{\theta}P_0^{-1}(x_0 - \hat{x}_0) &= 0 \\ P_0\lambda_0 - x_0 + \hat{x}_0 &= 0 \\ x_0 &= \hat{x}_0 + P_0\lambda_0 \end{aligned} \tag{11.48}$$

From the second expression in Equation (11.47) we obtain

$$\lambda_N = 0 \tag{11.49}$$

From the third expression in Equation (11.47) we obtain

$$\begin{aligned} -\frac{2}{\theta}Q_k^{-1}w_k + \frac{2}{\theta}\lambda_{k+1} &= 0 \\ w_k &= Q_k\lambda_{k+1} \end{aligned} \tag{11.50}$$

This can be substituted into the process dynamics equation to obtain

$$x_{k+1} = F_k x_k + Q_k\lambda_{k+1} \tag{11.51}$$

From the fourth expression in Equation (11.47) we obtain

$$\begin{aligned} \frac{2\lambda_k}{\theta} &= 2\bar{S}_k(x_k - \hat{x}_k) + \frac{2}{\theta}H_k^T R_k^{-1}(y_k - H_k x_k) + \frac{2}{\theta}F_k^T\lambda_{k+1} \\ \lambda_k &= F_k^T\lambda_{k+1} + \theta\bar{S}_k(x_k - \hat{x}_k) + H_k^T R_k^{-1}(y_k - H_k x_k) \end{aligned} \tag{11.52}$$

At this point we have to make an assumption in order to proceed any further. From Equation (11.48) we know that $x_0 = \hat{x}_0 + P_0\lambda_0$, so we will assume that

$$x_k = \mu_k + P_k\lambda_k \tag{11.53}$$

for all k, where μ_k and P_k are some functions to be determined, with P_0 given, and the initial condition $\mu_0 = \hat{x}_0$. That is, we assume that x_k is an affine function of λ_k. This assumption may or may not turn out to be valid. We will proceed as if the assumption were true, and if our results turn out to be correct then we will know that our assumption was indeed valid. Substituting Equation (11.53) into Equation (11.51) gives

$$\mu_{k+1} + P_{k+1}\lambda_{k+1} = F_k\mu_k + F_k P_k\lambda_k + Q_k\lambda_{k+1} \tag{11.54}$$

Substituting Equation (11.53) into Equation (11.52) gives

$$\lambda_k = F_k^T\lambda_{k+1} + \theta\bar{S}_k(\mu_k + P_k\lambda_k - \hat{x}_k) + H_k^T R_k^{-1}\left[y_k - H_k(\mu_k + P_k\lambda_k)\right] \tag{11.55}$$

Rearranging this equation gives

$$\begin{aligned} &\lambda_k - \theta\bar{S}_k P_k\lambda_k + H_k^T R_k^{-1}H_k P_k\lambda_k = \\ &\qquad F_k^T\lambda_{k+1} + \theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^T R_k^{-1}(y_k - H_k\mu_k) \end{aligned} \tag{11.56}$$

This can be solved for λ_k as

$$\begin{aligned} \lambda_k &= \left[I - \theta\bar{S}_k P_k + H_k^T R_k^{-1}H_k P_k\right]^{-1} \times \\ &\quad \left[F_k^T\lambda_{k+1} + \theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^T R_k^{-1}(y_k - H_k\mu_k)\right] \end{aligned} \tag{11.57}$$

Substituting this expression for λ_k into Equation (11.54) gives

$$\begin{aligned} &\mu_{k+1} + P_{k+1}\lambda_{k+1} = F_k\mu_k + F_k P_k\left[I - \theta\bar{S}_k P_k + H_k^T R_k^{-1}H_k P_k\right]^{-1} \times \\ &\qquad \left[F_k^T\lambda_{k+1} + \theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^T R_k^{-1}(y_k - H_k\mu_k)\right] + Q_k\lambda_{k+1} \end{aligned} \tag{11.58}$$

This equation can be rearranged as follows:

$$\begin{aligned} &\mu_{k+1} - F_k\mu_k - F_k P_k\left[I - \theta\bar{S}_k P_k + H_k^T R_k^{-1}H_k P_k\right]^{-1} \times \\ &\left[\theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^T R_k^{-1}(y_k - H_k\mu_k)\right] = \\ &\Big[-P_{k+1} + F_k P_k\left[I - \theta\bar{S}_k P_k + H_k^T R_k^{-1}H_k P_k\right]^{-1}F_k^T + Q_k\Big]\lambda_{k+1} \end{aligned} \tag{11.59}$$

This equation is satisfied if both sides are zero. Setting the left side of the above equation equal to zero gives

$$\begin{aligned}\mu_{k+1} &= F_k\mu_k + F_kP_k\left[I - \theta\bar{S}_kP_k + H_k^TR_k^{-1}H_kP_k\right]^{-1}\times \\ &\quad \left[\theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^TR_k^{-1}(y_k - H_k\mu_k)\right] \qquad (11.60)\end{aligned}$$

with the initial condition

$$\mu_0 = \hat{x}_0 \qquad (11.61)$$

Setting the right side of Equation (11.59) equal to zero gives

$$\begin{aligned}P_{k+1} &= F_kP_k\left[I - \theta\bar{S}_kP_k + H_k^TR_k^{-1}H_kP_k\right]^{-1}F_k^T + Q_k \\ &= F_k\tilde{P}_kF_k^T + Q_k \qquad (11.62)\end{aligned}$$

where $\tilde{P}_k$ is defined by the above equation. That is,

$$\begin{aligned}\tilde{P}_k &= P_k\left[I - \theta\bar{S}_kP_k + H_k^TR_k^{-1}H_kP_k\right]^{-1} \\ &= \left[P_k^{-1} - \theta\bar{S}_k + H_k^TR_k^{-1}H_k\right]^{-1} \qquad (11.63)\end{aligned}$$

From the above equation we see that if P_k, $\bar{S}_k$, and R_k are symmetric, then $\tilde{P}_k$ will be symmetric. We see from Equation (11.62) that if Q_k is also symmetric, then P_{k+1} will be symmetric. So if P_0, Q_k, R_k, and S_k are symmetric for all k, then $\tilde{P}_k$ and P_k will be symmetric for all k. The values of x_0 and w_k that provide a stationary point of J can be summarized as follows:

$$\begin{aligned}x_0 &= \hat{x}_0 + P_0\lambda_0 \\ w_k &= Q_k\lambda_{k+1} \\ \lambda_N &= 0 \\ \lambda_k &= \left[I - \theta\bar{S}_kP_k + H_k^TR_k^{-1}H_kP_k\right]^{-1}\times \\ &\quad \left[F_k^T\lambda_{k+1} + \theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^TR_k^{-1}(y_k - H_k\mu_k)\right] \\ P_{k+1} &= F_kP_k\left[I - \theta\bar{S}_kP_k + H_k^TR_k^{-1}H_kP_k\right]^{-1}F_k^T + Q_k \\ \mu_0 &= \hat{x}_0 \\ \mu_{k+1} &= F_k\mu_k + F_kP_k\left[I - \theta\bar{S}_kP_k + H_k^TR_k^{-1}H_kP_k\right]^{-1}\times \\ &\quad \left[\theta\bar{S}_k(\mu_k - \hat{x}_k) + H_k^TR_k^{-1}(y_k - H_k\mu_k)\right] \qquad (11.64)\end{aligned}$$

The fact that we were able to find a stationary point of J shows that we were correct in our assumption that x_k was an affine function of λ_k. In the following section, given these values of x_0 and w_k, we will find the values of $\hat{x}_k$ and y_k that provide a stationary point of J.

11.3.2 Stationarity with respect to $\hat{x}$ and y

The problem in this section is to find a stationary point (with respect to $\hat{x}_k$ and y_k) of $J = \psi(x_k)|_{k=0} + \sum_{k=0}^{N-1}\mathcal{L}_k$ (subject to the constraint $x_{k+1} = F_kx_k + w_k$). This problem is solved given the fact that x_0 and w_k have already been set to their maximizing values as described in Section 11.3.1. From Equation (11.53), and the initial condition of μ_k in Equation (11.61), we see that

$$\begin{aligned} \lambda_k &= P_k^{-1}(x_k - \mu_k) \\ \lambda_0 &= P_0^{-1}(x_0 - \hat{x}_0) \end{aligned} \tag{11.65}$$

We therefore obtain

$$\begin{aligned} ||\lambda_0||_{P_0}^2 &= \lambda_0^T P_0 \lambda_0 \\ &= (x_0 - \hat{x}_0)^T P_0^{-T} P_0 P_0^{-1}(x_0 - \hat{x}_0) \\ &= (x_0 - \hat{x}_0)^T P_0^{-1}(x_0 - \hat{x}_0) \\ &= ||x_0 - \hat{x}_0||_{P_0^{-1}}^2 \end{aligned} \tag{11.66}$$

Therefore, Equation (11.45) becomes

$$J = \frac{-1}{\theta}||\lambda_0||_{P_0}^2 + \sum_{k=0}^{N-1}\left[||x_k - \hat{x}_k||_{\bar{S}_k}^2 - \frac{1}{\theta}\left(||w_k||_{Q_k^{-1}}^2 + ||y_k - H_k x_k||_{R_k^{-1}}^2\right)\right] \tag{11.67}$$

Substituting for x_k from Equation (11.53) in this expression gives

$$J = \frac{-1}{\theta}||\lambda_0||_{P_0}^2 + \tag{11.68}$$
$$\sum_{k=0}^{N-1}\left[||\mu_k + P_k\lambda_k - \hat{x}_k||_{\bar{S}_k}^2 - \frac{1}{\theta}\left(||w_k||_{Q_k^{-1}}^2 + ||y_k - H_k(\mu_k + P_k\lambda_k)||_{R_k^{-1}}^2\right)\right]$$

Consider the term $w_k^T Q_k^{-1} w_k$ in the above equation. Substituting for w_k from Equation (11.50) in this term gives

$$\begin{aligned} w_k^T Q_k^{-1} w_k &= \lambda_{k+1}^T Q_k^T Q_k^{-1} Q_k \lambda_{k+1} \\ &= \lambda_{k+1}^T Q_k \lambda_{k+1} \end{aligned} \tag{11.69}$$

where we have used the fact that Q_k is symmetric. Equation (11.68) can therefore be written as

$$J = \frac{-1}{\theta}||\lambda_0||_{P_0}^2 + \tag{11.70}$$
$$\sum_{k=0}^{N-1}\left[||\mu_k + P_k\lambda_k - \hat{x}_k||_{\bar{S}_k}^2 - \frac{1}{\theta}||y_k - H_k(\mu_k + P_k\lambda_k)||_{R_k^{-1}}^2\right] - \frac{1}{\theta}\sum_{k=0}^{N-1}||\lambda_{k+1}||_{Q_k}^2$$

Now we take a slight digression to notice that

$$\sum_{k=0}^{N}\lambda_k^T P_k \lambda_k - \sum_{k=0}^{N-1}\lambda_k^T P_k \lambda_k = 0 \tag{11.71}$$

The reason that this equation is correct is because from Equation (11.49) we know that $\lambda_N = 0$. Therefore, the last term in the first summation above is equal to zero and the two summations are equal. The above equation can be written as

$$\begin{aligned}
0 &= \lambda_0^T P_0 \lambda_0 + \sum_{k=1}^{N} \lambda_k^T P_k \lambda_k - \sum_{k=0}^{N-1} \lambda_k^T P_k \lambda_k \\
&= \lambda_0^T P_0 \lambda_0 + \sum_{k=0}^{N-1} \lambda_{k+1}^T P_{k+1} \lambda_{k+1} - \sum_{k=0}^{N-1} \lambda_k^T P_k \lambda_k \\
&= \frac{-1}{\theta} ||\lambda_0||_{P_0}^2 - \frac{1}{\theta} \sum_{k=0}^{N-1} (\lambda_{k+1}^T P_{k+1} \lambda_{k+1} - \lambda_k^T P_k \lambda_k) \qquad (11.72)
\end{aligned}$$

We can subtract this zero term to the cost function of Equation (11.70) to obtain

$$\begin{aligned}
J &= \sum_{k=0}^{N-1} \Big[||\mu_k + P_k\lambda_k - \hat{x}_k||_{\bar{S}_k}^2 - \\
&\quad \frac{1}{\theta}||\lambda_{k+1}||_{Q_k}^2 + \frac{1}{\theta}(\lambda_{k+1}^T P_{k+1}\lambda_{k+1} - \lambda_k^T P_k \lambda_k) - \frac{1}{\theta}||y_k - H_k(\mu_k + P_k\lambda_k)||_{R_k^{-1}}^2 \Big] \\
&= \sum_{k=0}^{N-1} \Big[(\mu_k - \hat{x}_k)^T \bar{S}_k (\mu_k - \hat{x}_k) + 2(\mu_k - \hat{x}_k)^T \bar{S}_k P_k \lambda_k + \lambda_k^T P_k \bar{S}_k P_k \lambda_k + \\
&\quad \frac{1}{\theta}\lambda_{k+1}^T (P_{k+1} - Q_k)\lambda_{k+1} - \frac{1}{\theta}\lambda_k^T P_k \lambda_k - \frac{1}{\theta}(y_k - H_k\mu_k)^T R_k^{-1} (y_k - H_k\mu_k) + \\
&\quad \frac{2}{\theta}(y_k - H_k\mu_k)^T R_k^{-1} H_k P_k \lambda_k - \frac{1}{\theta}\lambda_k^T P_k H_k^T R_k^{-1} H_k P_k \lambda_k \Big] \qquad (11.73)
\end{aligned}$$

Now we consider the term $\lambda_{k+1}^T(P_{k+1} - Q_k)\lambda_{k+1}$ in the above expression. Substituting for P_{k+1} from Equation (11.62) in this term gives

$$\begin{aligned}
\lambda_{k+1}^T(P_{k+1} - Q_k)\lambda_{k+1} &= \lambda_{k+1}^T(Q_k + F_k\tilde{P}_k F_k^T - Q_k)\lambda_{k+1} \\
&= \lambda_{k+1}^T F_k \tilde{P}_k F_k^T \lambda_{k+1} \qquad (11.74)
\end{aligned}$$

But from Equation (11.55) we see that

$$F_k^T \lambda_{k+1} = \lambda_k - \theta\bar{S}_k(\mu_k + P_k\lambda_k - \hat{x}_k) - H_k^T R_k^{-1} [y_k - H_k(\mu_k + P_k\lambda_k)] \qquad (11.75)$$

Substituting this expression for $F_k^T\lambda_{k+1}$ into Equation (11.74) gives

$$\begin{aligned}
&\lambda_{k+1}^T(P_{k+1} - Q_k)\lambda_{k+1} \\
&= \left\{\lambda_k - \theta\bar{S}_k(\mu_k + P_k\lambda_k - \hat{x}_k) - H_k^T R_k^{-1}[y_k - H_k(\mu_k + P_k\lambda_k)]\right\}^T \\
&\quad \tilde{P}_k \left\{\lambda_k - \theta\bar{S}_k(\mu_k + P_k\lambda_k - \hat{x}_k) - H_k^T R_k^{-1}[y_k - H_k(\mu_k + P_k\lambda_k)]\right\} \\
&= \left\{\lambda_k^T(I - \theta P_k\bar{S}_k + P_k H_k^T R_k^{-1} H_k) - \theta(\mu_k - \hat{x}_k)^T\bar{S}_k - \right. \\
&\quad \left. (y_k - H_k\mu_k)^T R_k^{-1} H_k\right\} \tilde{P}_k \left\{\lambda_k^T(I - \theta P_k\bar{S}_k + P_k H_k^T R_k^{-1} H_k) - \right. \\
&\quad \left. \theta(\mu_k - \hat{x}_k)^T\bar{S}_k - (y_k - H_k\mu_k)^T R_k^{-1} H_k\right\}^T \qquad (11.76)
\end{aligned}$$

Now note from Equation (11.63) that $(I - \theta P_k\bar{S}_k + P_k H_k^T R_k^{-1} H_k) = P_k\tilde{P}_k^{-1}$. Making this substitution in the above equation gives the following.

$$
\begin{aligned}
&\lambda_{k+1}^T(P_{k+1}-Q_k)\lambda_{k+1} \\
&= \left\{\lambda_k^T P_k \tilde{P}_k^{-1} - \theta(\mu_k-\hat{x}_k)^T\bar{S}_k - (y_k - H_k\mu_k)^T R_k^{-1} H_k\right\} \\
&\quad \tilde{P}_k \left\{\lambda_k^T P_k \tilde{P}_k^{-1} - \theta(\mu_k-\hat{x}_k)^T\bar{S}_k - (y_k - H_k\mu_k)^T R_k^{-1} H_k\right\}^T \\
&= \lambda_k^T P_k \tilde{P}_k^{-1} P_k \lambda_k - \theta(\mu_k-\hat{x}_k)^T \bar{S}_k P_k \lambda_k - (y_k - H_k\mu_k)^T R_k^{-1} H_k P_k \lambda_k - \\
&\quad \theta\lambda_k P_k \bar{S}_k(\mu_k-\hat{x}_k) + \theta^2(\mu_k-\hat{x}_k)^T\bar{S}_k\tilde{P}_k\bar{S}_k(\mu_k-\hat{x}_k) + \\
&\quad \theta(y_k-H_k\mu_k)^T R_k^{-1} H_k \tilde{P}_k \bar{S}_k(\mu_k-\hat{x}_k) - \lambda_k^T P_k H_k^T R_k^{-1}(y_k - H_k\mu_k) + \\
&\quad \theta(\mu_k-\hat{x}_k)^T\bar{S}_k\tilde{P}_k H_k^T R_k^{-1}(y_k-H_k\mu_k) + \\
&\quad (y_k-H_k\mu_k)^T R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1}(y_k-H_k\mu_k)
\end{aligned} \tag{11.77}
$$

Notice that the above expression is a scalar. That means that each term on the right side is a scalar, which means that each term is equal to its transpose. For example, consider the second term on the right side. Since it is a scalar, we see that $\theta(\mu_k-\hat{x}_k)^T\bar{S}_k P_k\lambda_k = \theta\lambda_k^T P_k\bar{S}_k(\mu_k-\hat{x}_k)$. (We have used the fact that P_k and $\bar{S}_k$ are symmetric, and θ is a scalar.) Equation (11.77) can therefore be written as

$$
\begin{aligned}
&\lambda_{k+1}^T(P_{k+1}-Q_k)\lambda_{k+1} \\
&= \lambda_k^T P_k \tilde{P}_k^{-1} P_k \lambda_k - 2\theta(\mu_k-\hat{x}_k)^T \bar{S}_k P_k \lambda_k - \\
&\quad 2(y_k - H_k\mu_k)^T R_k^{-1} H_k P_k \lambda_k + \theta^2(\mu_k-\hat{x}_k)^T\bar{S}_k\tilde{P}_k\bar{S}_k(\mu_k-\hat{x}_k) + \\
&\quad 2\theta(\mu_k-\hat{x}_k)^T\bar{S}_k\tilde{P}_k H_k^T R_k^{-1}(y_k-H_k\mu_k) + \\
&\quad (y_k-H_k\mu_k)^T R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1}(y_k-H_k\mu_k)
\end{aligned} \tag{11.78}
$$

Now note from Equation (11.63) that

$$
\begin{aligned}
\tilde{P}_k^{-1} &= \left[I - \theta\bar{S}_k P_k + H_k^T R_k^{-1} H_k P_k\right] P_k^{-1} \\
&= P_k^{-1}\left[P_k^{-1} - \theta\bar{S}_k + H_k^T R_k^{-1} H_k\right] P_k^{-1} \\
&= P_k^{-1}\left[I - P_k\theta\bar{S}_k + P_k H_k^T R_k^{-1} H_k\right]
\end{aligned} \tag{11.79}
$$

We therefore see that

$$
\begin{aligned}
\lambda_k^T P_k \tilde{P}_k^{-1} P_k \lambda_k &= \lambda_k^T\left[I - \theta P_k\bar{S}_k + P_k H_k^T R_k^{-1} H_k\right] P_k \lambda_k \\
&= \lambda_k^T P_k \lambda_k - \theta\lambda_k^T P_k \bar{S}_k P_k \lambda_k + \lambda_k^T P_k H_k^T R_k^{-1} H_k P_k \lambda_k
\end{aligned} \tag{11.80}
$$

Substituting this into Equation (11.78) gives

$$
\begin{aligned}
&\lambda_{k+1}^T(P_{k+1}-Q_k)\lambda_{k+1} \\
&= \lambda_k^T P_k \lambda_k - \theta\lambda_k^T P_k \bar{S}_k P_k \lambda_k + \lambda_k^T P_k H_k^T R_k^{-1} H_k P_k \lambda_k - \\
&\quad 2\theta(\mu_k-\hat{x}_k)^T\bar{S}_k P_k\lambda_k - 2(y_k-H_k\mu_k)^T R_k^{-1} H_k P_k \lambda_k + \\
&\quad \theta^2(\mu_k-\hat{x}_k)^T\bar{S}_k\tilde{P}_k\bar{S}_k(\mu_k-\hat{x}_k) + 2\theta(\mu_k-\hat{x}_k)^T\bar{S}_k\tilde{P}_k H_k^T R_k^{-1}(y_k-H_k\mu_k) + \\
&\quad (y_k-H_k\mu_k)^T R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1}(y_k-H_k\mu_k)
\end{aligned} \tag{11.81}
$$

Substituting this equation for $\lambda_{k+1}^T(P_{k+1}-Q_k)\lambda_{k+1}$ into Equation (11.73) gives the following.

$$
\begin{aligned}
J &= \sum_{k=0}^{N-1} \Big[(\mu_k - \hat{x}_k)^T \bar{S}_k (\mu_k - \hat{x}_k) - \frac{1}{\theta}(y_k - H_k\mu_k)^T R_k^{-1}(y_k - H_k\mu_k) + \\
&\quad \theta(\mu_k - \hat{x})_k^T \bar{S}_k \tilde{P}_k \bar{S}_k (\mu_k - \hat{x}_k) + 2(\mu_k - \hat{x}_k)^T \bar{S}_k \tilde{P}_k H_k^T R_k^{-1}(y_k - H_k\mu_k) + \\
&\quad \frac{1}{\theta}(y_k - H_k\mu_k)^T R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1}(y_k - H_k\mu_k) \Big] \\
&= \sum_{k=0}^{N-1} \Big[(\mu_k - \hat{x}_k)^T (\bar{S}_k + \theta \bar{S}_k \tilde{P}_k \bar{S}_k)(\mu_k - \hat{x}_k) + \\
&\quad 2(\mu_k - \hat{x}_k)^T \bar{S}_k \tilde{P}_k H_k^T R_k^{-1}(y_k - H_k\mu_k) + \\
&\quad \frac{1}{\theta}(y_k - H_k\mu_k)^T (R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1} - R_k^{-1})(y_k - H_k\mu_k) \Big]
\end{aligned} \tag{11.82}
$$

Now recall our original objective: we are trying to find the stationary point of J with respect to $\hat{x}_k$ and y_k. If we take the partial derivative of the above expression for J with respect to $\hat{x}_k$ and y_k and set them equal to 0 we obtain

$$
\begin{aligned}
\frac{\partial J}{\partial \hat{x}_k} &= 2(\bar{S}_k + \theta \bar{S}_k \tilde{P}_k \bar{S}_k)(\hat{x}_k - \mu_k) + 2\bar{S}_k \tilde{P}_k H_k^T R_k^{-1}(H_k\mu_k - y_k) \\
&= 0 \\
\frac{\partial J}{\partial y_k} &= \frac{2}{\theta}(R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1} - R_k^{-1})(y_k - H_k\mu_k) + 2R_k^{-1} H_k \tilde{P}_k \bar{S}_k (\mu_k - \hat{x}_k) \\
&= 0
\end{aligned} \tag{11.83}
$$

These equations are clearly satisfied for the following values of $\hat{x}_k$ and y_k:

$$
\begin{aligned}
\hat{x}_k &= \mu_k \\
y_k &= H_k\mu_k
\end{aligned} \tag{11.84}
$$

These are the extremizing values of $\hat{x}_k$ and y_k. However, we still are not sure if these extremizing values give a local minimum or maximum of J. Recall that the second derivative of J tells us what kind of stationary point we have. If the second derivative is positive definite, then our stationary point is a minimum. If the second derivative is negative definite, then our stationary point is a maximum. If the second derivative has both positive and negative eigenvalues, then our stationary point is a saddle point. The second derivative of J with respect to $\hat{x}_k$ can be computed as

$$
\frac{\partial^2 J}{\partial \hat{x}_k^2} = 2(\bar{S}_k + \theta \bar{S}_k \tilde{P}_k \bar{S}_k) \tag{11.85}
$$

Our $\hat{x}_k$ will therefore be a minimizing value of J if $(\bar{S}_k + \theta \bar{S}_k \tilde{P}_k \bar{S}_k)$ is positive definite. The value of S_k chosen for use in Equation (11.36) should always be positive definite, which means that $\bar{S}_k$ defined in Equation (11.44) will be positive definite. This means that our $\hat{x}_k$ will be a minimizing value of J if $\tilde{P}_k$ is positive definite.

So, from the definition of $\tilde{P}_k$ in Equation (11.63), the condition required for $\hat{x}_k$ to minimize J is that $\left(P_k^{-1} - \theta \bar{S}_k + H_k^T R_k^{-1} H_k\right)^{-1}$ be positive definite. This is equivalent to requiring that $(P_k^{-1} - \theta\bar{S}_k + H_k^T R_k^{-1} H_k)$ be positive definite. The individual terms in this expression are always positive definite [note in particular from Equation (11.62) that P_k will be positive definite if $\tilde{P}_k$ is positive definite]. So the condition for $\hat{x}_k$ to minimize J is that $\theta\bar{S}_k$ be "small enough" so that $(P_k^{-1} - \theta\bar{S}_k + H_k^T R_k^{-1} H_k)$ is positive definite. Requiring that $\theta\bar{S}_k$ be small can be accomplished three different ways.

1. $\theta\bar{S}_k$ will be small if θ is small. This means that the performance requirement specified in Equation (11.37) is not too stringent. As long as our performance requirement is not too stringent then the problem will have a solution. If, however, the performance requirement is too stringent (i.e., θ is large) then the problem will not have a solution.

2. $\theta\bar{S}_k$ will be small if L_k is small. This statement is based on the relationship between $\bar{S}_k$ and L_k as shown in Equation (11.44). From Equation (11.36) we see that the numerator of the cost function is given as $(x_k - \hat{x}_k)^T L_k^T S_k L_k (x_k - \hat{x}_k)$. So if L_k is small we see that the numerator of the cost function will be small, which means that it will be easier to minimize the cost function. If, however, L_k is too large, then the problem will not have a solution.

3. $\theta\bar{S}_k$ will be small if S_k is small. This statement is based on the relationship between $\bar{S}_k$ and S_k as shown in Equation (11.44). From Equation (11.36) we see that the numerator of the cost function is given as $(x_k - \hat{x}_k)^T L_k^T S_k L_k (x_k - \hat{x}_k)$. So if S_k is small we see that the numerator of the cost function will be small, which means that it will be easier to minimize the cost function. If, however, S_k is too large, then the problem will not have a solution.

Note from Equation (11.62) that the positive definiteness of $\tilde{P}_k$ implies the positive definiteness of P_{k+1}. Therefore, if P_0 is positive definite (per our original problem statement), and $\tilde{P}_k$ is positive definite for all k, then P_k will also be positive definite for all k.

It is also academically interesting (though of questionable utility) to note the conditions under which the y_k that we found in Equation (11.84) will be a maximizing value of J. (Recall that y_k is chosen by nature, our adversary, to maximize the cost function.) The second derivative of J with respect to y_k can be computed as

$$\begin{aligned} \frac{\partial^2 J}{\partial y_k^2} &= \frac{2}{\theta}(R_k^{-1} H_k \tilde{P}_k H_k^T R_k^{-1} - R_k^{-1}) \\ &= \frac{2}{\theta} R_k^{-1}(H_k \tilde{P}_k H_k^T - R_k) R_k^{-1} \end{aligned} \tag{11.86}$$

R_k and R_k^{-1}, specified by the user as part of the problem statement in Equation (11.36), should always be positive definite. So the second derivative above will be negative definite (which means that y_k will be a maximizing value of J) if $(R_k - H_k \tilde{P}_k H_k^T)$ is positive definite. This requirement can be satisfied in two ways.

1. $(R_k - H_k \tilde{P}_k H_k^T)$ will be positive definite if R_k is large enough. A large value of R_k means that the denominator of the cost function of Equation (11.36) will be small, which means that the cost function will be large. A large cost function value is easier to maximize and will therefore tend to have a

maximizing value for y_k. Also note that the designer typically chooses R_k to be proportional to the magnitude of the measurement noise. If the user knows that the measurement noise is large, then R_k will be large, which again will result in a problem with a maximizing value for y_k. In other words, nature will be better able to maximize the cost function if the measurement noise is large.

2. $(R_k - H_k \tilde{P}_k H_k^T)$ will be positive definite if H_k is small enough. If H_k becomes smaller, that means that the measurement noise becomes larger relative to the size of the measurements, as seen in Equation (11.34). In other words, a small value of H_k means a smaller signal-to-noise ratio for the measurements. A small signal-to-noise ratio gives nature a better opportunity to find a maximizing value of y_k.

Of course, we are not really interested in finding a maximizing value of y_k. Our goal was to find the minimizing value of x_k. The H_∞ filter algorithm can be summarized as follows.

The discrete-time H_∞ filter

1. The system equations are given as

$$\begin{aligned} x_{k+1} &= F_k x_k + w_k \\ y_k &= H_k x_k + v_k \\ z_k &= L_k x_k \end{aligned} \tag{11.87}$$

   where w_k and v_k are noise terms, and our goal is to estimate z_k.

2. The cost function is given as

$$J_1 = \frac{\sum_{k=0}^{N-1} ||z_k - \hat{z}_k||_{S_k}^2}{||x_0 - \hat{x}_0||_{P_0^{-1}}^2 + \sum_{k=0}^{N-1}\left(||w_k||_{Q_k^{-1}}^2 + ||v_k||_{R_k^{-1}}^2\right)} \tag{11.88}$$

   where P_0, Q_k, R_k, and S_k are symmetric, positive definite matrices chosen by the engineer based on the specific problem.

3. The cost function can be made to be less than $1/\theta$ (a user-specified bound) with the following estimation strategy, which is derived from Equations (11.44), (11.60), (11.62), and (11.84):

$$\begin{aligned} \bar{S}_k &= L_k^T S_k L_k \\ K_k &= P_k \left[I - \theta \bar{S}_k P_k + H_k^T R_k^{-1} H_k P_k\right]^{-1} H_k^T R_k^{-1} \\ \hat{x}_{k+1} &= F_k \hat{x}_k + F_k K_k (y_k - H_k \hat{x}_k) \\ P_{k+1} &= F_k P_k \left[I - \theta \bar{S}_k P_k + H_k^T R_k^{-1} H_k P_k\right]^{-1} F_k^T + Q_k \end{aligned} \tag{11.89}$$

4. The following condition must hold at each time step k in order for the above estimator to be a solution to the problem:

$$P_k^{-1} - \theta \bar{S}_k + H_k^T R_k^{-1} H_k > 0 \tag{11.90}$$

11.3.3 A comparison of the Kalman and H_∞ filters

Comparing the Kalman filter in Equation (11.12) and the H_∞ filter in Equation (11.89) reveals some fascinating connections. For instance, in the H_∞ filter, Q_k, R_k, and P_0 are design parameters chosen by the user based on *a priori* knowledge of the magnitude of the process disturbance w_k, the measurement disturbance v_k, and the initial estimation error $(x_0 - \hat{x}_0)$. In the Kalman filter, w_k, v_k, and $(x_0 - \hat{x}_0)$ are zero-mean, and Q_k, R_k, and P_0 are their respective covariances.

Now suppose we use $L_k = S_k = I$ in the H_∞ filter. That is, we are interested in estimating the entire state, and we want to weight all of the estimation errors equally in the cost function. If we use $\theta = 0$ then the H_∞ filter reduces to the Kalman filter (assuming Q_k, R_k, and P_0 are chosen as above). This provides an interesting interpretation of the Kalman filter; that is, the Kalman filter is the minimax filter in the case that the performance bound in Equation (11.36) is set equal to ∞. We see that although the Kalman filter minimizes the variance of the estimation error (as discussed in Section 5.2), it does not provide any guarantee as far as limiting the worst-case estimation error. That is, it does not guarantee any bound for the cost function of Equation (11.36).

The Kalman and H_∞ filter equations have an interesting difference. If we want to estimate a linear combination of states using the Kalman filter, the estimator is the same regardless of the linear combination that we want to estimate. That is, if we want to estimate $L_k x_k$ using the Kalman filter, the answer is the same regardless of the L_k matrix that we choose. However, in the H_∞ approach, the resulting filter depends strongly on L_k and the particular linear combination of states that we want to estimate.

Note that the H_∞ filter of Equation (11.89) is identical to the Kalman filter except for subtraction of the term $\theta \bar{S}_k P_k$ in the K_k and P_{k+1} equations. Recall from Section 5.5 that the Kalman filter can be made more robust to unmodeled noise and unmodeled dynamics by artificially increasing Q_k in the Kalman filter equations. This results in a larger covariance P_k, which in turn results in a larger gain K_k. From Equation (11.89) we can see that subtracting $\theta \bar{S}_k P_k$ on the right side of the P_{k+1} equation tends to make P_{k+1} larger (since the subtraction is inside a matrix inverse operation). Similarly, subtracting $\theta \bar{S}_k P_k$ on the right side of the K_k equation tends to make K_k larger. Increasing Q_k in the Kalman filter is conceptually the same as increasing P_k and K_k. Therefore, the H_∞ filter equations make intuitive sense when compared with the Kalman filter equations. The H_∞ filter is a worst-case filter in the sense that it assumes that w_k, v_k, and x_0 will be chosen by nature to maximize the cost function. The H_∞ filter is therefore robust by design. Comparing the H_∞ filter with the Kalman filter, we can see that the H_∞ filter is simply a robust version of the Kalman filter. When we robustified the Kalman filter in Section 5.5 to add tolerance to unmodeled noise and dynamics, we did not derive an optimal way to increase Q_k. However, H_∞ filter theory shows us the optimal way to robustify the Kalman filter.

11.3.4 Steady-state H_∞ filtering

If the underlying system and the design parameters are time-invariant, then it may be possible to obtain a steady-state solution to the H_∞ filtering problem. Suppose that our system is given as

$$\begin{aligned} x_{k+1} &= Fx_k + w_k \\ y_k &= Hx_k + v_k \\ z_k &= Lx_k \end{aligned} \tag{11.91}$$

where w_k and v_k are noise terms. Our goal is to estimate z_k such that

$$\lim_{N\to\infty} \frac{\sum_{k=0}^{N-1} ||z_k - \hat{z}_k||_S^2}{\sum_{k=0}^{N-1}\left(||w_k||_{Q^{-1}}^2 + ||v_k||_{R^{-1}}^2\right)} < \frac{1}{\theta} \tag{11.92}$$

where Q, R, and S are symmetric positive definite matrices chosen by the engineer based on the specific problem. The steady-state filter of Equation (11.89) becomes

$$\begin{aligned} \bar{S} &= L^T S L \\ K &= P\left[I - \theta\bar{S}P + H^T R^{-1} H P\right]^{-1} H^T R^{-1} \\ \hat{x}_{k+1} &= F\hat{x}_k + FK_k(y_k - H\hat{x}_k) \\ P &= FP\left[I - \theta\bar{S}P + H^T R^{-1} H P\right]^{-1} F^T + Q \end{aligned} \tag{11.93}$$

The following condition must hold in order for the above estimator to be a solution to the problem:

$$P^{-1} - \theta\bar{S} + H^T R^{-1} H > 0 \tag{11.94}$$

If θ, L, R, or S is too large, or if H is too small, then the H_∞ estimator will not have a solution. Note that the expression for P in Equation (11.93) can be written as

$$P = F\left[P^{-1} - \theta\bar{S} + H^T R^{-1} H\right]^{-1} F^T + Q \tag{11.95}$$

Applying the matrix inversion lemma to the inverse in the above expression gives

$$\begin{aligned} P &= F\left\{P - P\left[(H^T R^{-1} H - \theta\bar{S})^{-1} + P\right]^{-1} P\right\} F^T + Q \\ &= FPF^T - FP\left[(H^T R^{-1} H - \theta\bar{S})^{-1} + P\right]^{-1} PF^T + Q \end{aligned} \tag{11.96}$$

This is a discrete-time algebraic Riccati equation that can be solved with control system software.[2] If control system software is not available, then the algebraic Riccati equation can be solved by numerically iterating the discrete-time Riccati equation of Equation (11.89) until it converges to a steady-state value. The steady-state filter is much easier to implement in a system in which real-time computational effort or code size is a serious consideration. The disadvantage of the steady-state filter is that (theoretically) it does not perform as well as the time-varying filter. However, the reduced performance that is seen in the steady-state filter is often a small fraction of the optimal performance, whereas the computational savings can be significant.

[2]For example, in MATLAB's Control System Toolbox we can use the command DARE($F^T, I, Q, (H^T R^{-1} H - \theta\bar{S})^{-1}$.

EXAMPLE 11.2

Suppose we are trying to estimate a randomly varying scalar on the basis of noisy measurements. We have the scalar system

$$\begin{aligned} x_{k+1} &= x_k + w_k \\ y_k &= x_k + v_k \\ z_k &= x_k \end{aligned} \tag{11.97}$$

This system could describe our attempt to estimate a noisy voltage. The voltage is essentially constant, but it is subject to random fluctuations, hence the noise term w_k in the process equation. Our measurement of the voltage is also subject to noise or instrument bias, hence the noise term v_k in the measurement equation. We see in this example that $F = H = L = 1$. Further suppose that $Q = R = S = 1$ in the cost function of Equation (11.88). Then the discrete-time Riccati equation associated with the H_∞ filter equations becomes

$$\begin{aligned} P_{k+1} &= F_k P_k \left[I - \theta \bar{S}_k P_k + H_k^T R_k^{-1} H_k P_k\right]^{-1} F_k^T + Q_k \\ &= P_k \left[1 - \theta P_k + P_k\right]^{-1} + 1 \end{aligned} \tag{11.98}$$

This can be solved numerically or analytically as a function of time for a given θ to give P_k, and then the H_∞ gain can be obtained as

$$\begin{aligned} K_k &= P_k \left[I - \theta \bar{S}_k P_k + H_k^T R_k^{-1} H_k P_k\right]^{-1} H_k^T R_k^{-1} \\ &= P_k \left[1 - \theta P_k + P_k\right]^{-1} \end{aligned} \tag{11.99}$$

We can set $P_{k+1} = P_k$ in Equation (11.98) to obtain the steady-state solution for P_k. This gives

$$\begin{aligned} P &= P(1 - \theta P + P)^{-1} + 1 \\ P(1 - \theta P + P) &= P + (1 - \theta P + P) \\ (1-\theta)P^2 + (\theta - 1)P - 1 &= 0 \\ P &= \frac{1 - \theta \pm \sqrt{(\theta-1)(\theta-5)}}{2(1-\theta)} \end{aligned} \tag{11.100}$$

As we discussed earlier, in order for this value of P to be a solution to the H_∞ estimation problem, P must be positive definite. The first solution for P is positive if $\theta < 1$, and both solutions for P are positive if $\theta \geq 5$. Another condition for the solution of the H_∞ estimation problem is that

$$\begin{aligned} P^{-1} - \theta \bar{S} + H^T R^{-1} H &> 0 \\ P^{-1} - \theta + 1 &> 0 \end{aligned} \tag{11.101}$$

If $\theta < 1$ then the first solution for P satisfies this bound. However, if $\theta \geq 5$, then neither solution for P satisfies this bound. Combining this data shows that the H_∞ estimator problem has a solution for $\theta < 1$. Every H_∞ estimator problem will have a solution for θ less than some upper bound because of the nature of the cost function.

For a general estimator gain K the estimate can be written as

$$\begin{aligned} \hat{x}_{k+1} &= F\hat{x}_k + FK(y_k - H_k\hat{x}_k) \\ &= (1-K)\hat{x}_k + Ky_k \end{aligned} \quad (11.102)$$

If we choose $\theta = 1/2$, then we obtain $P = 2$ and $K = 1$. As seen from the above equation, this results in $\hat{x}_{k+1} = y_k$. In other words, the estimator ignores the previous estimate and simply sets the estimate equal to the previous measurement. As θ increases toward 1, P increases above 2 and approaches ∞, and the estimator gain K increases greater than 1 and also approaches ∞. In this case, the estimator will actually place a negative weight on the previous estimate and compensate by placing additional weight on the measurement. If θ increases too much (gets too close to 1) then the estimator gain K will be greater than 2 and the H_∞ estimator will be unstable. It is always a good idea to check the stability of your H_∞ filter. If the filter is unstable then you should probably decrease θ to obtain a stable filter. As θ decreases below 1/2, P decreases below 2 and the gain K decreases below 1. In this case, the estimator balances the relative weight placed on the previous estimate and the measurement.

A Kalman filter to estimate x_k is equivalent to an H_∞ filter with $\theta = 0$. In this case, we obtain the positive definite solution of the steady-state Riccati equation as $P = (1+\sqrt{5})/2$. This gives a steady-state estimator gain $K = (1+\sqrt{5})/(3+\sqrt{5}) = (\sqrt{5}-1)/2 \approx 0.62$. The Kalman filter gain is smaller than the H_∞ filter gain for $\theta > 0$, which means that the Kalman filter relies less on measurements and more on the system model. The Kalman filter gives an optimal estimate if the model and the noise statistics are known, but it may undervalue the measurements if there are errors in the system model or the assumed noise statistics.

Figure 11.2 shows the true state x_k and the estimate $\hat{x}_k$ when the steady-state Kalman and H_∞ filters are used to estimate the state. The H_∞ filter was designed with $\theta = 1/3$, which gave a filter gain $K = (3+3\sqrt{7})/(8+2\sqrt{7}) \approx 0.82$. The disturbances w_k and v_k were both normally distributed zero-mean white noise sequences with standard deviations equal to 10. The performance of the two filters is very similar. The RMS estimation error of the Kalman filter is 3.6 and the RMS estimation error of the H_∞ filter is 4.1. As expected, the Kalman filter performs better than the H_∞ filter. However, suppose that the process noise has a mean of 10. Figure 11.3 shows the performance of the filters for this situation. In this case the H_∞ filter performs better. The RMS estimation error of the Kalman filter is 15.6 and the RMS estimation error of the H_∞ filter is 12.0.

If we choose $\theta = 1/10$ then we obtain $P = 5/3$ and $K = 2/3$. As θ gets smaller, the H_∞ estimator gain gets closer and closer to the Kalman filter gain.

$\nabla\nabla\nabla$

11.3.5 The transfer function bound of the H_∞ filter

In this section, we show that the steady-state H_∞ filter derived in the previous section bounds the transfer function from the noise to the estimation error, if Q,

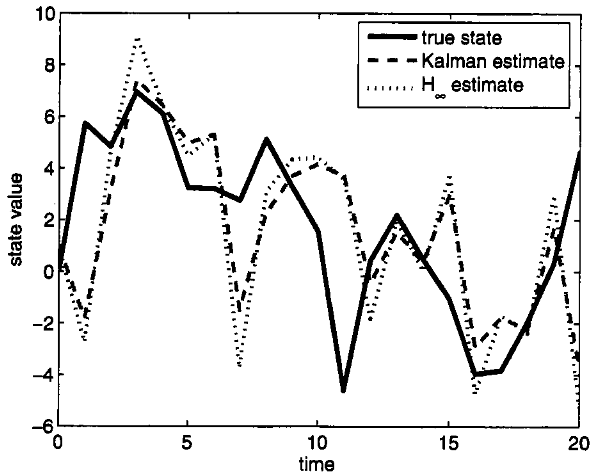

Figure 11.2 Example 11.2 results. Kalman and H$_\infty$ filter peformance when the noise statistics are known. The Kalman gain is 0.62 and the H$_\infty$ gain is 0.82. The Kalman filter performs about 12% better than the H$_\infty$ filter.

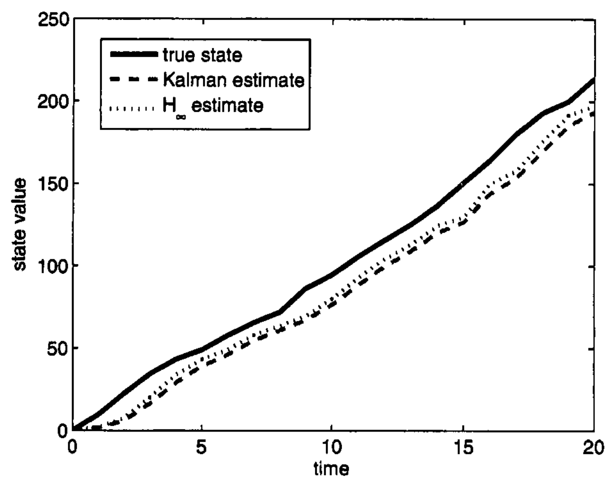

Figure 11.3 Example 11.2 results. Kalman and H$_\infty$ filter peformance when the process noise is biased. The Kalman gain is 0.62 and the H$_\infty$ gain is 0.82. The H$_\infty$ filter performs about 23% better than the Kalman filter.

R, and S are all identity matrices. Recall that the two-norm of a column vector x is defined as

$$||x||_2^2 = x^T x \tag{11.103}$$

Now suppose we have a time-varying vector $x_0, x_1, x_2, \cdots$. The signal two-norm of x is defined as

$$||x||_2^2 = \sum_{k=0}^{\infty} ||x_k||_2^2 \tag{11.104}$$

That is, the square of the signal two-norm is the sum of all of the squares of the vector two-norms that are taken at each time step.[3] Now suppose that we have a system with input u and output x, and the transfer function is $G(z)$. If the input u is comprised entirely of signals at the frequency ω and the sample time of the system is T, then we define the phase of u as $\phi = T\omega$. In this case the maximum gain from u to x is determined as

$$\sup_{u \neq 0} \frac{||x||_2}{||u||_2} = \sigma_1 \left[G\left(e^{j\phi}\right) \right] \tag{11.105}$$

where $\sigma_1(G)$ is the largest singular value of the matrix G. If u can be comprised of an arbitrary mix of frequencies, then the maximum gain from u to x is determined as follows:

$$\begin{aligned} \sup_{\phi} \frac{||x||_2}{||u||_2} &= \sup_{\phi} \sigma_1 \left[G\left(e^{j\phi}\right) \right] \\ &= ||G||_\infty \end{aligned} \tag{11.106}$$

The above equation defines $||G||_\infty$, which is the infinity-norm of the system that has the transfer function $G(z)$.[4]

Now consider Equation (11.92), the cost function that is bounded by the steady-state H_∞ filter:

$$J = \lim_{N \to \infty} \frac{\sum_{k=0}^{N-1} ||z_k - \hat{z}_k||_S^2}{\sum_{k=0}^{N-1} \left(||w_k||_{Q^{-1}}^2 + ||v_k||_{R^{-1}}^2 \right)} \tag{11.107}$$

If Q, R, and S are all equal to identity matrices, then

$$J = \lim_{N \to \infty} \frac{\sum_{k=0}^{N-1} ||z_k - \hat{z}_k||_2^2}{\sum_{k=0}^{N-1} \left(||w_k||_2^2 + ||v_k||_2^2 \right)} \tag{11.108}$$

Since the H_∞ filter makes this scalar less than $1/\theta$ for all w_k and v_k, we can write

$$\begin{aligned} ||G_{\tilde{z}e}||_\infty^2 &= \sup_{\phi} \frac{||z - \hat{z}||_2^2}{||w||_2^2 + ||v||_2^2} \\ &\leq \frac{1}{\theta} \end{aligned} \tag{11.109}$$

where we have defined $\tilde{z} = z - \hat{z}$, $e^T = \left[\begin{array}{cc} w^T & v^T \end{array} \right]^T$, and $G_{\tilde{z}e}$ is the system that has e as its input and $\tilde{z}$ as its output. We see that the steady-state H_∞ filter bounds the infinity-norm (i.e., the maximum gain) from the combined disturbances w and v to the estimation error $\tilde{z}$, if Q, R, and S are all identity matrices. Further information about the computation of infinity-norms and related issues can be found in [Bur99].

[3] Note that this definition means that many signals have unbounded signal two-norms. The signal two-norm can also be defined as the sum from $k = 0$ to a finite limit $k = N$.

[4] Note that the infinity-norm of a matrix has a definition that is different than the infinity-norm of a system. In general, the expression $||G||_\infty$ could refer either to the matrix infinity-norm or the system infinity-norm. The meaning needs to be inferred from the context unless it is explicitly stated.

EXAMPLE 11.3

Consider the system and filter discussed in Example 11.2:

$$\begin{aligned} x_{k+1} &= x_k + w_k \\ y_k &= x_k + v_k \\ \hat{x}_{k+1} &= (1-K)\hat{x}_k + K y_k \end{aligned} \tag{11.110}$$

The estimation error can be computed as

$$\begin{aligned} \tilde{x}_{k+1} &= x_{k+1} - \hat{x}_{k+1} \\ &= (1-K)\tilde{x}_k + w_k - K v_k \end{aligned} \tag{11.111}$$

Taking the z-transform of this equation gives

$$\begin{aligned} z\tilde{X}(z) &= (1-K)\tilde{X}(z) + W(z) - KV(z) \\ \tilde{X}(z) &= \frac{1}{z-1+K} \begin{bmatrix} 1 & -K \end{bmatrix} \begin{bmatrix} W(z) \\ V(z) \end{bmatrix} \\ &= G(z) \begin{bmatrix} W(z) \\ V(z) \end{bmatrix} \end{aligned} \tag{11.112}$$

$G(z)$, the transfer function from w_k and v_k to $\tilde{x}_k$, is a 2×1 matrix. This matrix has one singular value, which is computed as

$$\begin{aligned} \sigma^2(G) &= \lambda_{\max}\left[G(e^{j\phi})G^H(e^{j\phi})\right] \\ &= \frac{1+K^2}{(e^{j\phi}-1+K)(e^{-j\phi}-1+K)} \\ &= \frac{1+K^2}{K^2+2(K-1)(\cos\phi - 1)} \end{aligned} \tag{11.113}$$

The supremum of this expression occurs at $\phi = 0$ when $K \le 1$, so

$$\begin{aligned} \|G\|_\infty^2 &= \sup_\phi \sigma^2\left[G\left(e^{j\phi}\right)\right] \\ &= \frac{1+K^2}{K^2} \end{aligned} \tag{11.114}$$

Recall from Example 11.2 that $\theta = 1/2$ resulted in $K = 1$. In this case, the above expression indicates that $\|G\|_\infty^2 = 2 \le 1/\theta = 2$. In this case, the infinity-norm bound specified by θ is exact. Also recall from Example 11.2 that $\theta = 1/10$ resulted in $K = 2/3$. In this case, the above expression indicates that $\|G\|_\infty^2 = 13/4 \le 1/\theta = 10$. In this case, the infinity-norm bound specified by θ is quite conservative.

Note that as K increases, the infinity-norm from the noise to the estimation error decreases. However, the estimator also is unstable for $K > 1$. So even though large K reduces the infinity-norm of the estimator, it gives poor results. In other words, just because the effect of the noise on the estimation error is small does not necessarily prove that the estimator is good. For example, we could set the estimate $\hat{x}_k = \infty$ for all k. In that case, the noise